CMSC 2621
Lab Exercise 6

Due: See the assignment folder for the due date.

Overview:

In this exercise, you will create a potion brewing simulator where different ingredients can be mixed to create magical potions. You will use templates to handle various types of ingredients and exception handling to manage errors that may occur during the brewing process. By the end of this exercise, you will have a solid understanding of how templates and exception handling work in C++.

Requirements:

1. Download the PotionBrewer.cpp file that's linked in the D2L assignment. Follow the instructions in the comments and update only the locations in the code that have a "CHANGE_HERE"
2. Submit the source code and a document of a screenshot in a zip file to the D2L assignment. (Please see the instructions in the end of this document)

Submission Requirements:

You'll be submitting a zip file that contains the files below:

1. The source code file
2. A Word document containing screenshots of your source code and the full results of executing your program.

After creating a zip file containing all the required files, submit the zip file to the assignment folder on D2L.

Screenshot Requirements:

1. The screenshot(s) need to be clear enough to read the source code.
2. If the running results are too long to be captured, use multiple screenshots.
3. The running result(s) need to be captured in the screenshot(s) completely. If the running results are not complete, i.e. just partial results are captured in the screenshot(s), you may also receive a points deduction, depending on how complete the results in the screenshots are.

Assignment Grading:

1. If the program fails to be compiled using a compiler that supports C++ 20, a maximum grade of 40% will be given.
2. If any requirements are not met, for each unfulfillment, a corresponding penalty will be given.
3. Completeness of the lab assignment will also be considered.

Example Output:

```
Added ingredient: 5
Added ingredient: 10
Added ingredient: 20
Strength potion brewed! Effectiveness: 35
Added ingredient: 3.5
Added ingredient: 7.2
Healing potion brewed! Effectiveness: 10.7
Added ingredient: unicorn hair
Added ingredient: dragon scale
Mystical potion brewed! Ingredients: unicorn hair dragon scale
Added ingredient: poison
Error: Poison detected! The potion is ruined.
```